

CODE SPORT



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Over the next few columns, we will continue our discussion on data storage systems and look at how they are evolving to cater to the world of data-centric computing.

Last month, we had started our discussion of storage systems by looking at various concepts like SAN, NAS, etc. In this column, we will look at the concept of scale-up vs scale-out storage, and discuss their relative advantages and disadvantages.

Scale-up vs scale-out storage

Most of us are familiar with the concepts of scale-up and scale-out computing as they apply to server computing. A scale-up server is a single system with high CPU power and huge memory, and hence is able to handle increasing workloads. As the load on the system increases, it can be scaled up by adding more CPUs, more memory and storage to the single system, which is typically a shared memory system. The work is usually done by multiple threads and processes running on the same system, and they typically communicate through shared memory. This is a 'shared-everything' system. On the other hand, scale-out computing is when multiple nodes are part of the system, and they all act together to handle the workload, which is typically partitioned across the nodes. There is usually no shared memory across the nodes—they communicate through explicit messages over the network. Each node has both processing and storage elements. This is normally a 'shared-nothing' approach, where the nodes interact in a loosely coupled manner. The scale-out system can be scaled up to handle additional loads by adding more nodes to it; the work can be distributed across newly added nodes transparently, without the need to bring down the system.

Now, scale-out and scale-up storage systems have the same conceptual difference. In a scale-

up storage system, you add more and more storage capacity to the single storage node to meet increasing storage requirements. This can be achieved by adding many individual disk drives to a storage controller (or a pair of storage controllers so as to ensure failover and high availability). Since the storage capacity (i.e., number of individual disk drives) that a storage controller can support is limited, if storage requirements exceed that capacity, the only option is to move to the next bigger controller, with a higher capacity. With scale-out storage, this problem is avoided, since each node has its own storage controller, and as you add more nodes to meet increasing storage demands, it allows the controller architecture to grow as well. Scale-out storage is typically marketed under the slogan, "Pay as you grow", since it allows you to add nodes as and when your storage requirements increase, instead of having to over-provision from the beginning itself.

The next question that arises is about performance and cost. How do scale-up and scale-out storage systems compare in terms of performance? Before that, we need to understand what the common measures of performance for storage systems are. For computing systems, you can specify its performance in terms of its clock frequency (1/clock frequency gives the clock cycle period) and Cycles Per Instruction (CPI). It is possible to approximate the execution time for a program as (clock period * CPI * Instruction Count). For a computation system, the number of operations/instructions completed per second is a performance measure. This is typically specified as how many MIPS (million instructions per second) or how many